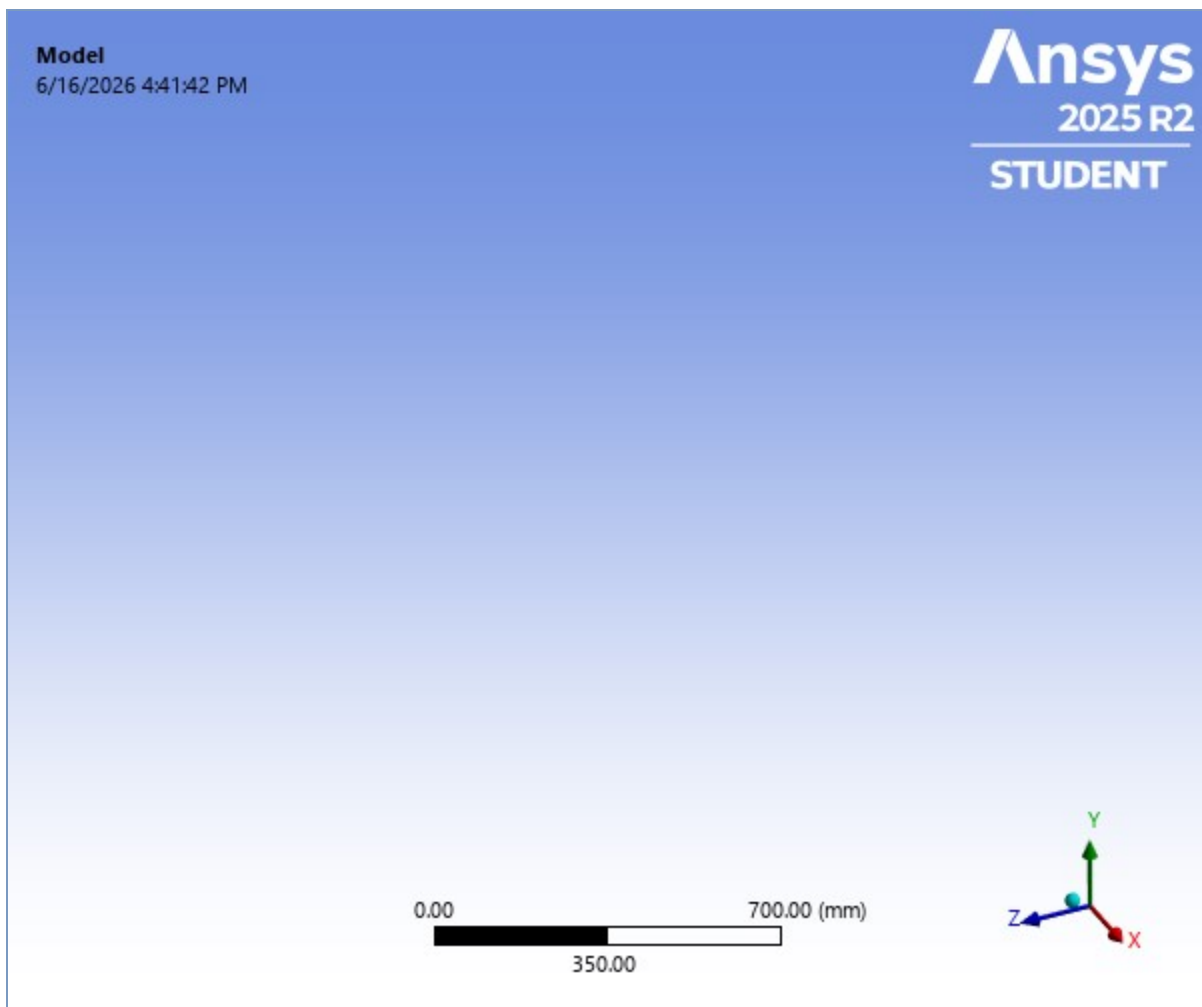




Project*

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Subject	FEA On Beam
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Product Version	2025 R2
Save Project Before Solution	No
Save Project After Solution	No



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Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

TABLE 2

Model (A4) > Geometry Imports

Object Name	<i>Geometry Imports</i>
State	Solved

TABLE 3

Model (A4) > Geometry Imports > Geometry Import (A3)

Object Name	<i>Geometry Import (A3)</i>
State	Solved
Definition	
Source	D:\Code Alpha\Task 2\FEA On Beam_files\dp0\SYS\DM\SYS.dsco
Type	Discovery
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes

Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4) > Geometry

Object Name	<i>Geometry</i>
State	Fully Defined
Definition	
Source	D:\Code Alpha\Task 2\FEA On Beam_files\dp0\SYS\DM\SYS.dsco
Type	Discovery
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	200. mm
Length Y	200. mm
Length Z	4000. mm
Properties	
Volume	1.6e+008 mm ³
Mass	1256. kg
Scale Factor Value	1.
Statistics	
Bodies	1
Active Bodies	1
Nodes	2952
Elements	468
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	

Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Parts

Object Name	<i>Beam\Boss-Extrude1</i>
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System
Reference Temperature	By Environment
Treatment	None
Material	
Assignment	Structural Steel
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	
Length X	200. mm
Length Y	200. mm
Length Z	4000. mm
Properties	
Volume	1.6e+008 mm ³
Mass	1256. kg
Centroid X	0. mm

Centroid Y	4.5175e-016 mm
Centroid Z	5.7824e-014 mm
Moment of Inertia Ip1	1.6789e+009 kg·mm ²
Moment of Inertia Ip2	1.6789e+009 kg·mm ²
Moment of Inertia Ip3	8.3733e+006 kg·mm ²
Statistics	
Nodes	2952
Elements	468
Mesh Metric	None
CAD Attributes	
PartTolerance:	0.00000001
Color:143.149.175	

TABLE 6
Model (A4) > Materials

Object Name	<i>Materials</i>
State	Fully Defined
Statistics	
Materials	1
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4) > Coordinate Systems > Coordinate System

Object Name	<i>Global Coordinate System</i>
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. mm
Origin Y	0. mm
Origin Z	0. mm
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]
Transfer Properties	
Source	
Read Only	No

Mesh

TABLE 8
Model (A4) > Mesh

Object Name	<i>Mesh</i>
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical

Element Order	Program Controlled
Element Size	Default
Sizing	
Use Adaptive Sizing	Yes
Resolution	5
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	4010.0 mm
Average Surface Area	5.4667e+005 mm ²
Minimum Edge Length	200.0 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Auto-Map Fillets	No
Automatic Methods	
Sheet Body Method	Prime Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	2952
Elements	468
Show Detailed Statistics	No

Static Structural (A5)

TABLE 9
Model (A4) > Analysis

Object Name	Static Structural (A5)
State	Solved
Definition	

Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 10
Model (A4) > Static Structural (A5) > Analysis Settings

Object Name	<i>Analysis Settings</i>
State	Fully Defined
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No

Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	D:\Code Alpha\Task 2\FEA On Beam_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	No
Solver Units	Active System
Solver Unit System	nmm

TABLE 11
Model (A4) > Static Structural (A5) > Loads

Object Name	<i>Fixed Support</i>	<i>Displacement</i>	<i>Force</i>
State	Fully Defined		
Scope			
Scoping Method	Geometry Selection		
Geometry	1 Edge		1 Face
Definition			
Type	Fixed Support	Displacement	Force
Suppressed	No		
Define By		Components	Vector
Coordinate System		Global Coordinate System	
X Component		0. mm (ramped)	
Y Component		0. mm (ramped)	
Z Component		Free	
Applied By			Surface Effect
Magnitude			2.e+005 N (ramped)
Direction			Defined

FIGURE 1
Model (A4) > Static Structural (A5) > Displacement

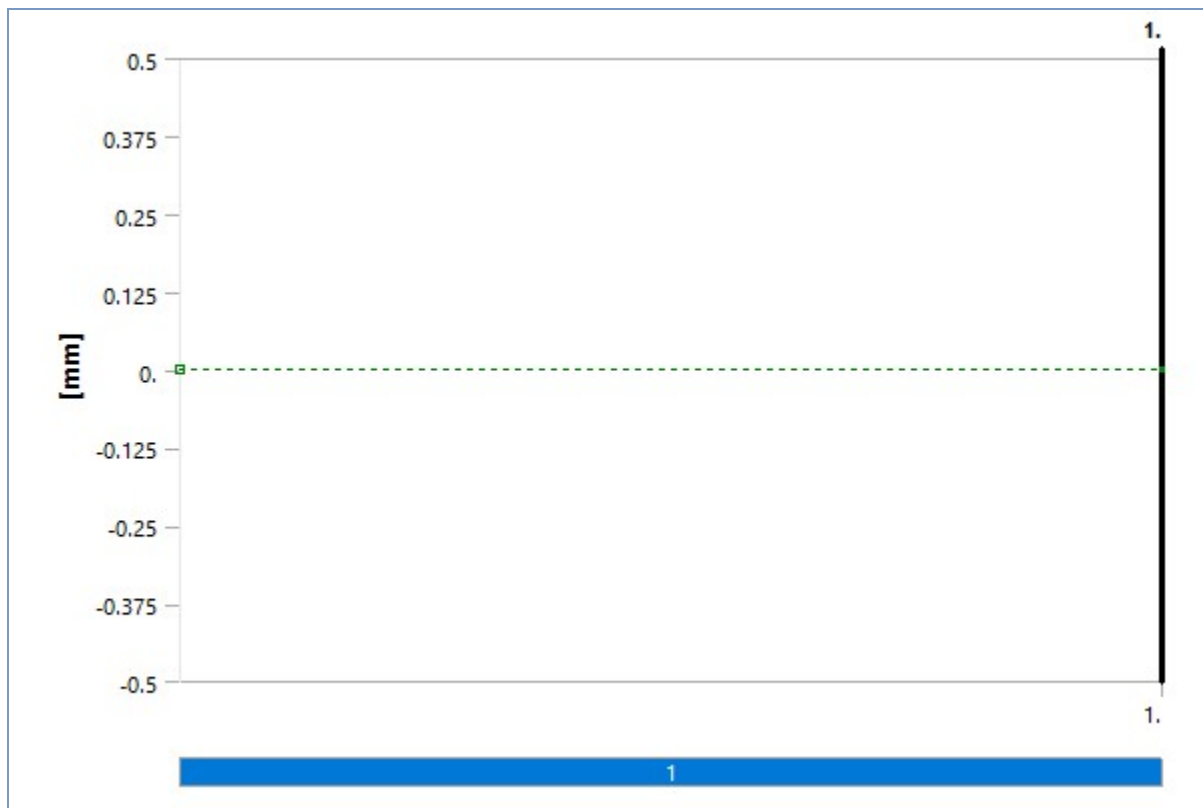
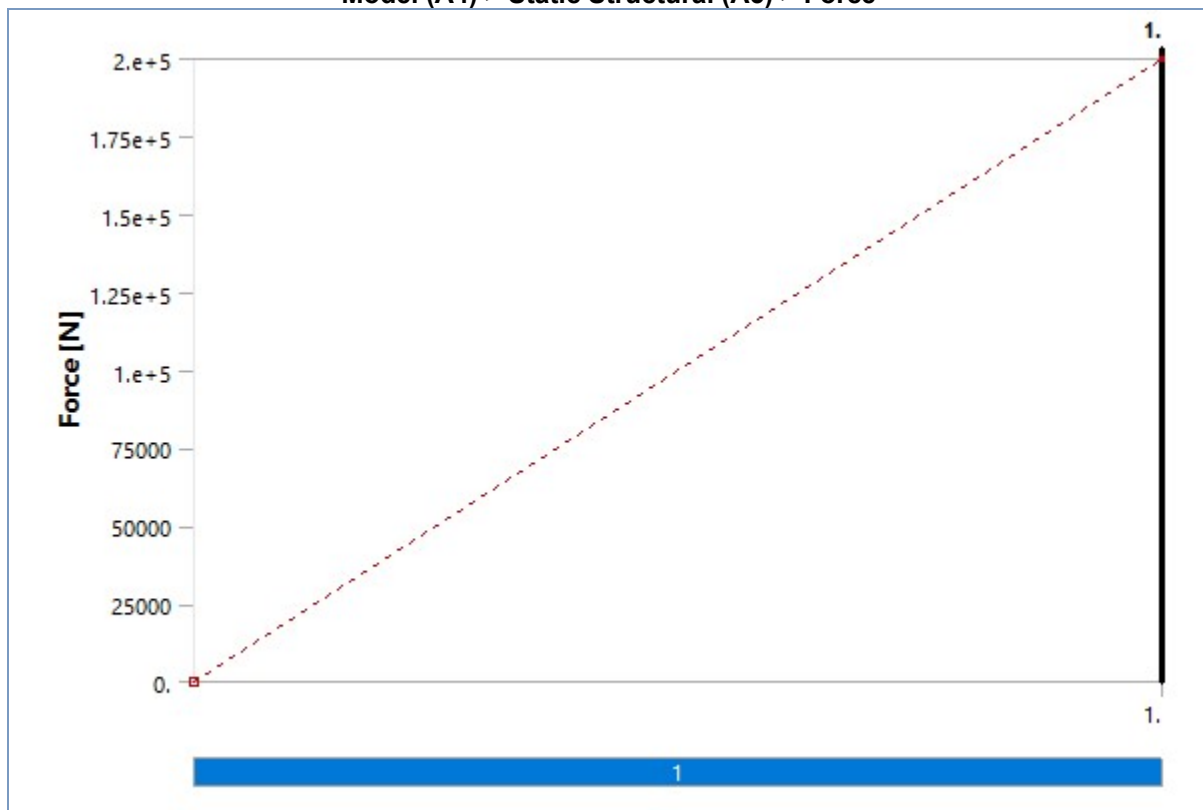


FIGURE 2
Model (A4) > Static Structural (A5) > Force



Solution (A6)

TABLE 12
Model (A4) > Static Structural (A5) > Solution

Object Name	<i>Solution (A6)</i>
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	18. s
MAPDL Memory Used	98. MB
MAPDL Result File Size	960. KB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 13
Model (A4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	<i>Solution Information</i>
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 14
Model (A4) > Static Structural (A5) > Solution (A6) > Results

Object Name	<i>Equivalent Stress</i>	<i>Directional Deformation</i>
State	Solved	
Scope		
Scoping Method	Geometry Selection	
Geometry	All Bodies	
Definition		
Type	Equivalent (von-Mises) Stress	Directional Deformation
By	Time	
Display Time	Last	
Separate Data by Entity	No	
Calculate Time History	Yes	
Identifier		
Suppressed	No	
Orientation		Y Axis
Coordinate System		Global Coordinate System
Integration Point Results		

Display Option	Averaged	
Average Across Bodies	No	
Results		
Minimum	0.12506 MPa	-6.3014 mm
Maximum	75.048 MPa	0. mm
Average	31.457 MPa	-3.9794 mm
Minimum Occurs On	Beam\Boss-Extrude1	
Maximum Occurs On	Beam\Boss-Extrude1	
Information		
Time	1. s	
Load Step	1	
Substep	1	
Iteration Number	1	

FIGURE 3
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

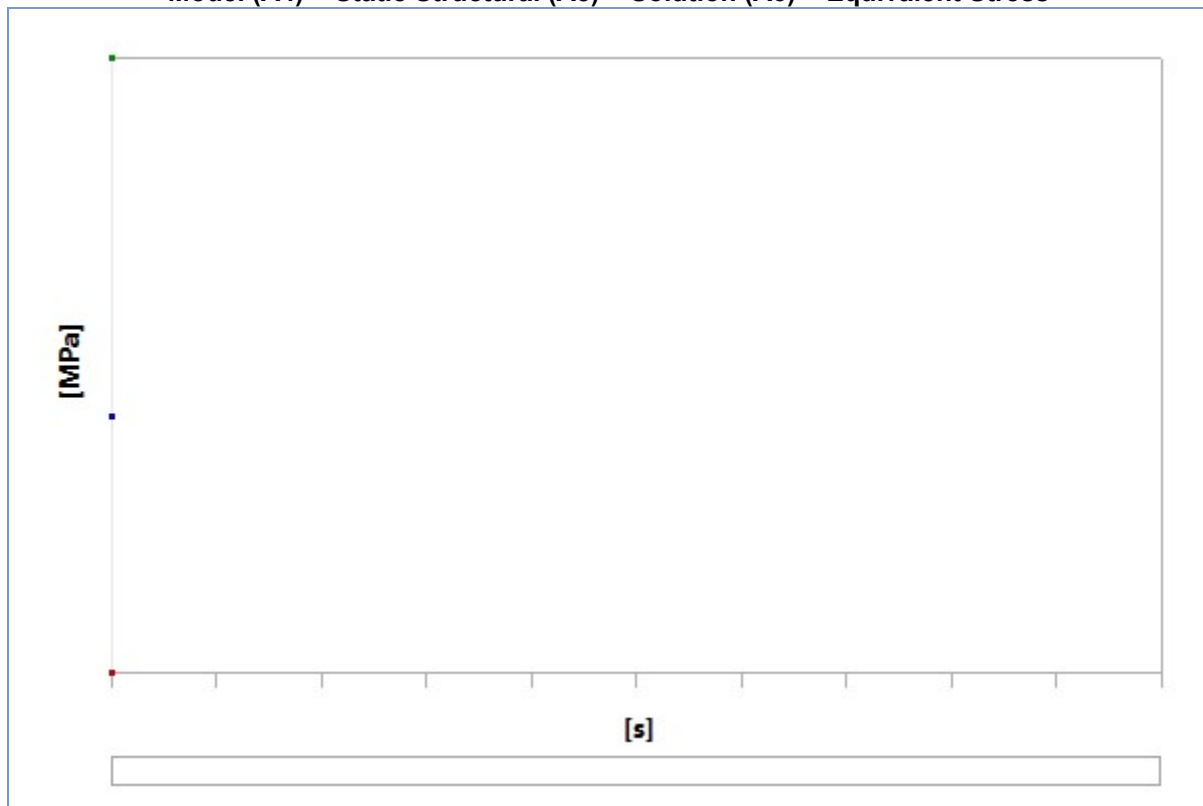
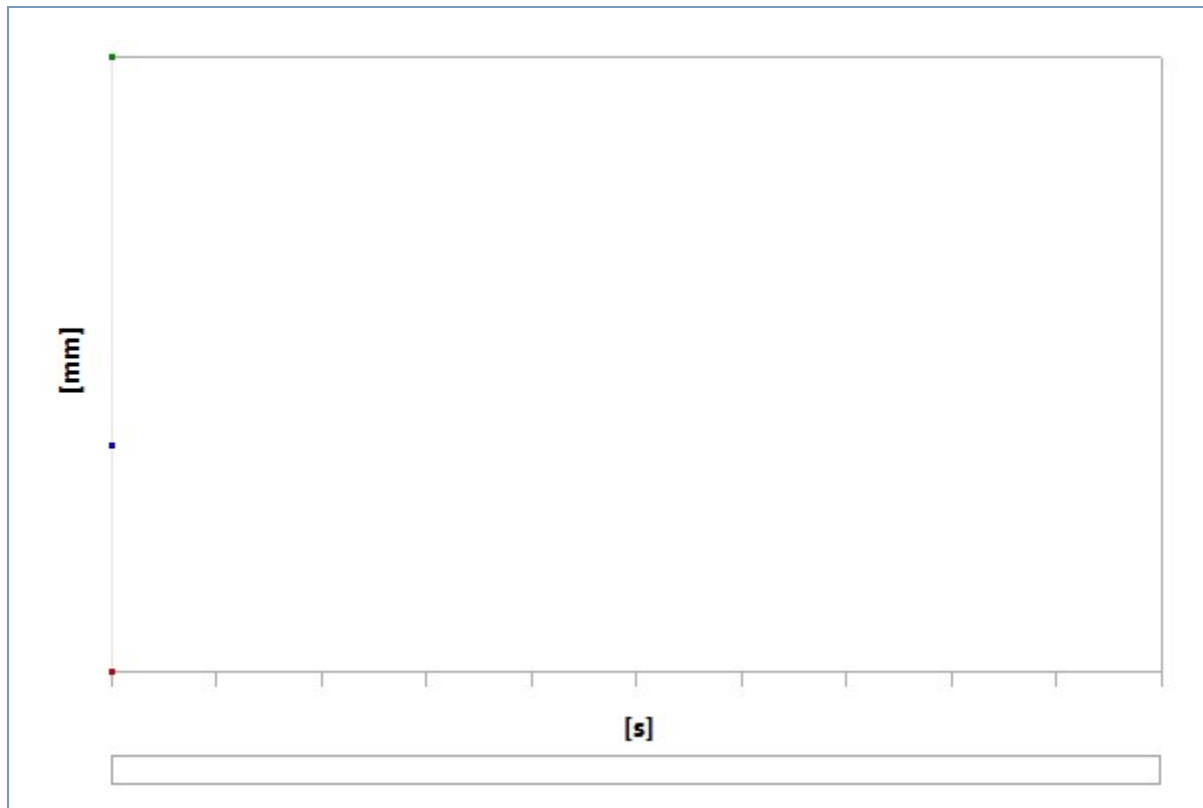


TABLE 15
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
1.	0.12506	75.048	31.457

FIGURE 4
Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation

**TABLE 16****Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation**

Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
1.	-6.3014	0.	-3.9794

TABLE 17**Model (A4) > Static Structural (A5) > Solution (A6) > Stress Safety Tools**

Object Name	<i>Stress Tool</i>
State	Solved
Definition	
Theory	Max Equivalent Stress
Stress Limit Type	Tensile Yield Per Material

TABLE 18**Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Results**

Object Name	<i>Safety Factor</i>
State	Solved
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Type	Safety Factor
By	Time
Display Time	Last
Separate Data by Entity	No
Calculate Time History	Yes
Identifier	
Suppressed	No

Integration Point Results	
Display Option	Averaged
Average Across Bodies	No
Results	
Minimum	3.3312
Minimum Occurs On	Beam\Boss-Extrude1
Information	
Time	1. s
Load Step	1
Substep	1
Iteration Number	1

FIGURE 5
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor

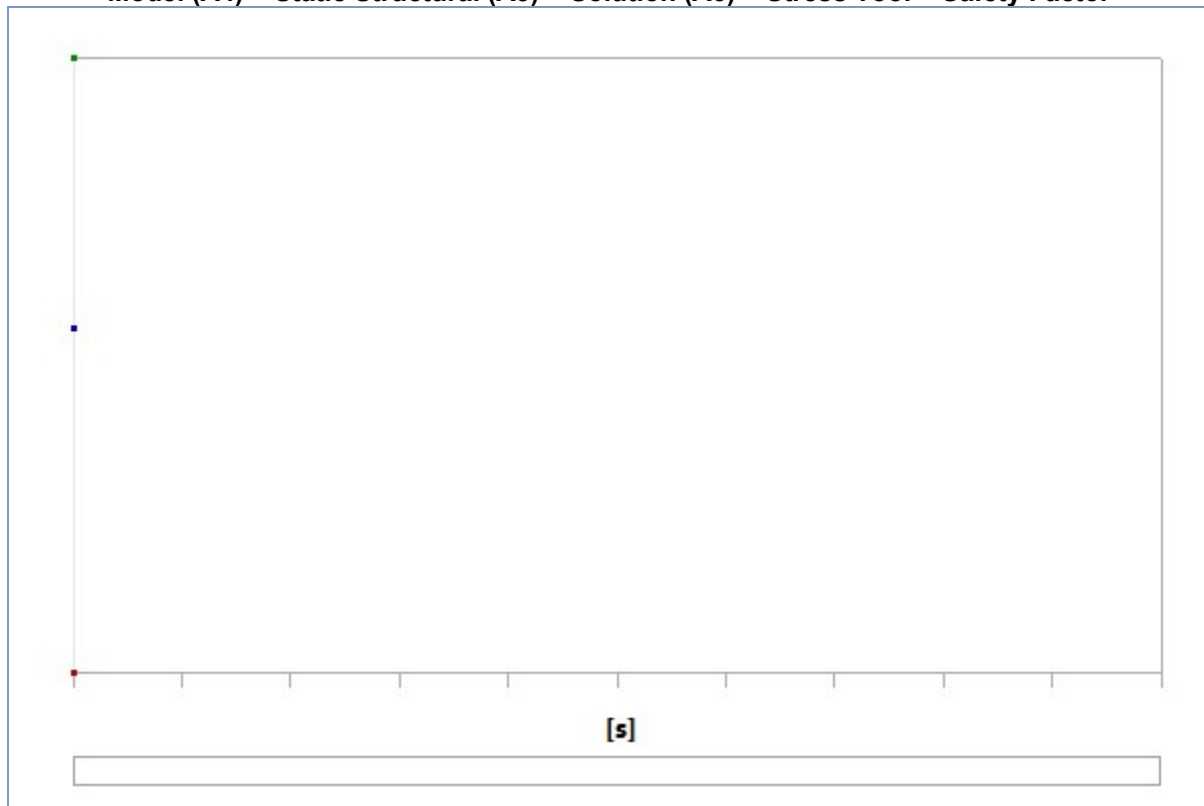


TABLE 19
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor

Time [s]	Minimum	Maximum	Average
1.	3.3312	15.	9.8739

Material Data

Structural Steel

TABLE 20
Structural Steel > Constants

Density	7.85e-006 kg mm ⁻³
Coefficient of Thermal Expansion	1.2e-005 K ⁻¹
Specific Heat	4.34e+005 mJ kg ⁻¹ K ⁻¹

Thermal Conductivity	6.05e-002 W mm ⁻¹ K ⁻¹
Resistivity	1.7e-004 ohm mm

TABLE 21
Structural Steel > Color

Red	Green	Blue
132	139	179

TABLE 22
Structural Steel > Compressive Ultimate Strength

Compressive Ultimate Strength MPa
0

TABLE 23
Structural Steel > Compressive Yield Strength

Compressive Yield Strength MPa
250

TABLE 24
Structural Steel > Tensile Yield Strength

Tensile Yield Strength MPa
250

TABLE 25
Structural Steel > Tensile Ultimate Strength

Tensile Ultimate Strength MPa
460

TABLE 26
Structural Steel > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature K
295.15

TABLE 27
Structural Steel > S-N Curve

Alternating Stress MPa	Cycles	Mean Stress MPa
3999	10	0
2827	20	0
1896	50	0
1413	100	0
1069	200	0
441	2000	0
262	10000	0
214	20000	0
138	1.e+005	0
114	2.e+005	0
86.2	1.e+006	0

TABLE 28
Structural Steel > Strain-Life Parameters

Strength Coefficient MPa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient MPa	Cyclic Strain Hardening Exponent
920	-0.106	0.213	-0.47	1000	0.2

TABLE 29
Structural Steel > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature K
2.e+005	0.3	1.6667e+005	76923	

TABLE 30
Structural Steel > Isotropic Relative Permeability

Relative Permeability
10000